

Slide 1: ACID Guarantees

23IT201 • Database Management Systems • Module 8: Transactions & Concurrency Control

Atomicity (all-or-nothing), Consistency (integrity constraints), Isolation (concurrency control), Durability (persisted logs).

Slide 2: Concurrency & Serializability

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Conflict serializable schedules produce outputs equivalent to some serial execution order.
Conflict operations: Read-Write, Write-Read, Write-Write.

Slide 3: Two-Phase Locking Protocols

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Growing Phase (acquiring locks) and Shrinking Phase (releasing locks). Strict 2PL holds exclusive locks until commit.